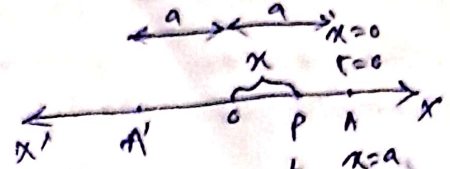


Simple Harmonic Motion

§ Definition:- A particle is said to execute a simple harmonic motion (S.H.M.), if it moves in a st. line such that its acceleration is always directed towards a fixed point in the line and is proportional to its distance from the fixed point.

§ Equation of motion of S.H.M:-

$$\begin{aligned} \ddot{x} &= -\mu x \\ \Rightarrow \frac{d^2x}{dt^2} &= -\mu (x - x') \\ \Rightarrow x &= a \cos \sqrt{\mu} t \end{aligned}$$



Periodic Time (T) = $\frac{2\pi}{\sqrt{\mu}}$

Frequency of oscillation (n) = $\frac{1}{T} = \frac{\sqrt{\mu}}{2\pi}$

Amplitude of S.H.M = a

§ General form:-

$$x = C \cos(\sqrt{\mu} t + \epsilon)$$

The quantity ϵ is called epoch and $(\sqrt{\mu} t + \epsilon)$ is called argument of the S.H.M.

Phase time = $t + \frac{\epsilon}{\sqrt{\mu}}$

§ Problem 1:- The speed v of a point moving along x-axis is given by $v = 16 - x$. Prove that the motion is simple harmonic. Find also amplitude.

Ans. Given that $v = 16 - x$

S.D. $\therefore$ to x

$$Q.D. \frac{dv}{dx} = -1$$

$$\therefore a = -1$$

Which indicates that the motion of the particle is simple harmonic motion.

Again

$$v = 16 - x$$

Putting $v=0$

$$16 - x = 0$$

$$\Rightarrow x = 16$$

$$\therefore \text{So amplitude} = \frac{16 - (-4)}{2} = 10$$

Problem 2:- If the velocity of a particle moving along the x-axis is given by $v = 86x - x^2 - 126$. Show that the motion is simple harmonic motion.

Ans. Given that $v = 86x - x^2 - 126$

$$\frac{dv}{dx} = 86 - 2x$$
$$a = \frac{dv}{dx} = 86 - 2x$$
$$\therefore \ddot{x} = -(x - 43)$$

Which indicates that the motion of the particle is a simple harmonic motion about the mean position $x = 43$.

Problem 3:- A particle moves along a st. line under the law of force given by $x = a \cos(\omega t + \phi)$. Show that the acceleration is directed towards the origin and varies as the distance.

Ans. Given that $x = a \cos(\omega t + \phi)$

$$\frac{dx}{dt} = -a\omega \sin(\omega t + \phi)$$
$$\ddot{x} = -a\omega^2 \cos(\omega t + \phi)$$
$$\therefore \ddot{x} = -\omega^2 x$$

Which shows that the acceleration is directed towards the origin and varies as the distance.

Problem 4:- A particle describes simple harmonic motion in a line with 2 cm as its amplitude. Its velocity when passing through the centre of oscillation is 12 cm/sec. Find the period.

Ans. For a particle executing S.H.M., the displacement x at any time t is given by

$$x = a \cos \sqrt{\mu} t$$

$$\text{S. D. } v = \frac{dx}{dt}$$

$$v = -a\sqrt{\mu} \sin \sqrt{\mu} t$$

$$= -a\sqrt{\mu} \sqrt{1 - \cos^2 \sqrt{\mu} t}$$

$$v = -a\sqrt{\mu} \sqrt{1 - \frac{x^2}{a^2}}$$

Here amplitude $(a) = 2 \text{ cm}$ and when $x=0$, $v=12 \frac{\text{cm}}{\text{s}}$

$$\therefore |v| = |-2\sqrt{\mu} \sqrt{1-0}| = 12$$

$$\text{or } 2\sqrt{\mu} = 12$$

$$(v \neq 0) \text{ or } \sqrt{\mu} = 6 \quad \frac{2\pi}{\sqrt{\mu}} = T \text{ or } \frac{2\pi}{6} = T$$

$$(v \neq 0) \text{ Period} = \frac{2\pi}{\sqrt{\mu}} = \frac{2\pi}{6} = \frac{\pi}{3} \text{ s}$$

3 Problem 5:- The distances of a particle, performing simple harmonic motion, from the middle point of its path at three consecutive seconds are observed to be x, y, z . Show that the time of a complete oscillation is $\frac{2\pi}{\cos^{-1}(\frac{x+z}{2y})}$.

Ans. Let the distances of the particle from the middle point of its path at $(t-1)$, t and $(t+1)$ be x, y and z respectively.

Now equation of motion of the particle is $x = a \cos \sqrt{\mu} t$

and a be the amplitude of the motion.

$$\text{Then we have } x = a \cos \sqrt{\mu} (t-1) \quad \text{--- (i)}$$

$$y = a \cos \sqrt{\mu} t \quad \text{--- (ii)}$$

$$z = a \cos \sqrt{\mu} (t+1) \quad \text{--- (iii)}$$

$$\text{Now } x+z = a \cos \sqrt{\mu} (t-1) + a \cos \sqrt{\mu} (t+1)$$

$$= 2a \cos \sqrt{\mu} t \cdot \cos \sqrt{\mu}$$

$$= 2y \cdot \cos \sqrt{\mu}$$

$$\therefore \cos \sqrt{\mu} = \frac{x+z}{2y} \Rightarrow \sqrt{\mu} = \cos^{-1} \left(\frac{x+z}{2y} \right)$$

$\therefore$ Time of a complete oscillation = Periodic time

$$T = \frac{2\pi}{\sqrt{\mu}} = \frac{2\pi}{\cos\left(\frac{\pi+2}{2\pi}\right)}$$

§ Problem 6:- In a simple harmonic motion, if f be the acceleration and v the velocity at any instant and T is the periodic time, then show that $f^2 T^2 + 4\pi^2 v^2$ is constant.

Ans. Let the equation be

$$\ddot{x} = -\mu x$$

and a be the amplitude of the motion.

$$\text{Then } f = -\mu x, T = \frac{2\pi}{\sqrt{\mu}} \text{ and } v = \mu(a^2 - x^2)$$

$$\begin{aligned} \text{Now } f^2 T^2 + 4\pi^2 v^2 &= \mu^2 x^2 \cdot \frac{4\pi^2}{\mu} + 4\pi^2 \cdot \mu(a^2 - x^2) \\ &= 4\pi^2 a^2 \mu \text{ which is constant.} \end{aligned}$$

§ Problem 7:- A particle of mass m moves along a straight line under an attraction $m n^2 x$ towards a point O on the line, where x is the distance from O . If $x=a$ and $\dot{x}=u$, when $t=0$, then show that the amplitude of motion is $\sqrt{a^2 + \frac{u^2}{n^2}}$.
In particular, if $u=0$, then find the period of oscillation.

Ans. The equation of motion is

$$m\ddot{x} = -mn^2 x$$

$$\text{or } \ddot{x} = -n^2 x \quad \text{--- (1)}$$

Multiplying both sides by $2\dot{x}$

$$2\dot{x}\ddot{x} = -n^2 x 2\dot{x}$$

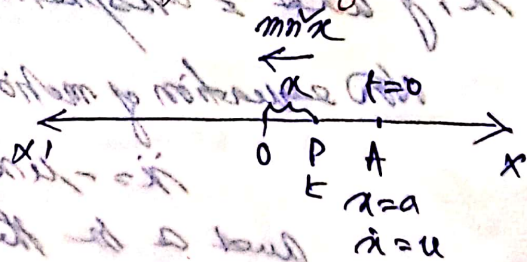
$$\text{or } \frac{d}{dt}(\dot{x}^2) = -2n^2 x \frac{dx}{dt}$$

Integrating w.r.t. t

$$\int \frac{d}{dt}(\dot{x}^2) dt = -2n^2 \int x \frac{dx}{dt} dt$$

$$\text{ie } \int \dot{x}^2 dt = -2n^2 \int x dx$$

$$\text{or } \dot{x}^2 = -2n^2 \frac{x^2}{2} + C, C = \text{D.C}$$



Initially at $t=0$, $x=a$, $\dot{x}=u$

$$u = \dot{x} = -\eta x + C$$

$$\Rightarrow C = u + \eta a$$

$$\therefore \dot{x} = -\eta x + u + \eta a$$

P

$$\ddot{x} + \eta \dot{x} = 0 \quad \text{--- (II)} \quad \left(\frac{u}{\eta} + \left(\frac{u}{\eta} + a \right) \frac{1}{\eta} = 0 \right)$$

General solution of (II) is

$$x = A \cos \eta t + B \sin \eta t \quad \text{--- (III)}$$

At $t=0$, $x=a$

$$\dot{x} = -A\eta \sin \eta t + B\eta \cos \eta t \quad \text{--- (IV)}$$

Initially at $t=0$, $x=a$, $\dot{x}=u$

$$\therefore \text{(III)} \Rightarrow a = A$$

$$\text{(IV)} \Rightarrow u = B\eta \Rightarrow B = \frac{u}{\eta}$$

$$\therefore \text{(III)} \Rightarrow x = a \cos \eta t + \frac{u}{\eta} \sin \eta t$$

$$\text{Let } a = r \cos \epsilon \quad \frac{u}{\eta} = -r \sin \epsilon = -r \sin \epsilon$$

So that $r = \sqrt{a^2 + \frac{u^2}{\eta^2}}$

$$\therefore x = r \cos(\eta t + \epsilon) \quad \text{where } r = \sqrt{a^2 + \frac{u^2}{\eta^2}} \quad \text{and } \epsilon = \tan^{-1} \left(-\frac{u}{\eta a} \right)$$

$$\text{where } r = \sqrt{a^2 + \frac{u^2}{\eta^2}} \quad \text{and } \epsilon = \tan^{-1} \left(-\frac{u}{\eta a} \right)$$

Now if $u=0$, then $r=a$ and $\frac{0}{\eta a} = 0$ and

Problem 8: A particle of mass m moves in a straight line under an attraction $m\eta x$ towards a point on the line where x is the distance from O . If $x=a$, $\dot{x}=u$ when $t=0$. Prove that the particle reaches O at time $\frac{1}{\eta} \cot^{-1} \left(-\frac{u}{\eta a} \right)$.

Ans. Equation of motion of the particle is $\ddot{x} + \eta x = 0$ --- (I)
and the solution under the given initial condition is
 $x = r \cos(\eta t + \epsilon)$ --- (II)
where $r = \sqrt{a^2 + \frac{u^2}{\eta^2}}$ and $\epsilon = -\frac{1}{\eta} \cot^{-1} \left(-\frac{u}{\eta a} \right)$.

If the particle reaches 0 at time t_1 , then from (1)

$$\cos(\pi t_1 + \epsilon) = 0$$

$$\Rightarrow \pi t_1 + \epsilon = \pi/2$$

$$\Rightarrow \pi t_1 = \pi/2 - \epsilon$$

$$= \pi/2 - \tan^{-1}\left(-\frac{u}{\pi a}\right)$$

$$\Rightarrow t_1 = \frac{1}{\pi} \cos^{-1}\left(-\frac{u}{\pi a}\right)$$

3) Problem 7:- A particle oscillates with simple harmonic motion of amplitude a and periodic time T . Find the expression of the Velocity in terms of

i) a , T and x

ii) a , t and T

$$\text{or } \int_0^T \frac{dx}{dt} = \frac{2\pi a}{T}$$

Ans. Let the equation of motion be

$$\ddot{x} = -\mu x \quad (\mu > 0)$$

and a be the amplitude.

Then $x = a \cos \omega t$

$$v = \mu (a^2 - x^2) \quad (1)$$

$$\text{And } T = \frac{2\pi}{\omega} \quad (2)$$

$$\therefore T^2 = \frac{4\pi^2}{\mu}$$

$$\Rightarrow \mu = \frac{4\pi^2}{T^2}$$

$$\therefore v = \mu (a^2 - x^2) = \frac{4\pi^2}{T^2} (a^2 - x^2)$$

$$\therefore v = \frac{2\pi}{T} \sqrt{a^2 - x^2}$$

$$(3) \text{ And } v = \frac{2\pi}{T} \sqrt{a^2 - a^2 \cos^2 \omega t} \quad \text{as } x = a \cos \omega t$$

$$v = \frac{2\pi}{T} a \sin \frac{2\pi}{T} t$$

$$\begin{aligned}
 \text{iv) } \int_0^T \ddot{v} dt &= \int_0^T \left(\frac{2\pi}{T} a \sin \frac{2\pi}{T} t \right) dt \\
 &= \frac{4\pi^2 a}{T^2} \int_0^T \sin^2 \frac{2\pi t}{T} dt \\
 &= \frac{2\pi^2 a}{T^2} \int_0^T \left(1 - \cos \frac{4\pi t}{T} \right) dt \\
 &= \frac{2\pi^2 a}{T^2} \left[t - \frac{T}{4\pi} \sin \frac{4\pi t}{T} \right]_0^T \\
 &= \frac{2\pi^2 a}{T^2} \left[T - \frac{T}{4\pi} \sin \frac{4\pi T}{T} \right]
 \end{aligned}$$

Since $\sin 4\pi = 0$.

§ Problem 11:- A particle moves along a st. line and its velocity v at a distance x from the origin is given by $v = 2 - x - x^2$. Show that the motion is simple harmonic.

Ans. Given that $v = 2 - x - x^2$

$$\begin{aligned}
 &\text{d. w. to } x \\
 &2v \frac{dv}{dx} = -1 - 2x \\
 &\text{or } \ddot{x} = -\frac{2}{2} \left(x + \frac{1}{2} \right) \\
 &\text{or } \ddot{x} = - \left(x + \frac{1}{2} \right)
 \end{aligned}$$

which shows that the motion of the particle executes S.H.M.

§ Problem 12:- A particle moving in a st. line has the following relation between the velocity v and the distance x from the origin

$$v = \sqrt{\mu - x^2} \quad (\mu > 0)$$

Show that the motion is S.H.M.

Ans. Given that $v = \sqrt{\mu - x^2}$

$$\begin{aligned}
 &\text{or } \ddot{v} = \mu - x^2 \\
 &\text{or } 2v \frac{dv}{dx} = -2x \\
 &\text{or } \boxed{\ddot{x} = -x}
 \end{aligned}$$

which indicates that the motion of the particle is S.H.M.

§ Problem 12:- The displacement of a moving particle at any time t is given by $x = a \cos kt + b \sin kt$.
Show that the particle executes simple harmonic motion.
Find the period of oscillation.

Ans. Given that $x = a \cos kt + b \sin kt$
D. w. r. to t

$$\dot{x} = -ak \sin kt + bk \cos kt$$

$$\text{D. w. r. to } t$$

$$\ddot{x} = -ak \cos kt - bk \sin kt$$

$$a \ddot{x} = -k^2 (a \cos kt + b \sin kt)$$

$$\text{or } \ddot{x} = -k^2 x$$

which indicates that the motion of the particle executes S.H.M. with period of oscillation is $\frac{2\pi}{k}$

$$x = a \cos kt + b \sin kt$$

$$x = a \cos kt + b \sin kt$$

$$x = a \cos kt + b \sin kt$$

$$(a + b) \frac{d}{dt} = \dot{x}$$

$$(a + b) = \dot{x}$$

which indicates that the motion of the particle

executes S.H.M.

$$x = a \cos kt + b \sin kt$$

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